

BoardMemory

API Contract Specification — Phase 1.5

Venture Foundry OS · Frozen Artifact Translation

1. API Design Principles

This API Contract Specification translates the frozen PRD (Functional and Non-Functional Requirements), Domain Model (Aggregates, Invariants, Domain Events) and Database Schema, together with the ten Core Services already approved in the System Architecture, into a technology-agnostic contract between the BoardMemory frontend and backend. Nothing below assumes a transport technology (REST, GraphQL, tRPC, an RPC framework) or an authentication protocol. Where a decision is not fully dictated by an approved artifact, it is derived from one of the six principles below and flagged inline, consistent with the flagging discipline already established across Phases 1.1–1.4.

- AC1 — Consistency. Every operation across all ten Core Services follows the same operation-specification shape — Purpose, Business Preconditions, Inputs, Outputs, Validation Rules, Business Errors, Success Conditions, Idempotency Expectations (Section 3) — and the same logical error model (Section 7). A frontend engineer never learns a second contract shape for a different service.
- AC2 — Traceability. Every operation carries forward NFR-01's acting-Person-and-timestamp requirement as an explicit contract obligation: no write operation accepts a request without an ActorContext (Section 4), and every response representing a Decision Graph fact echoes back who captured it and when, mirroring the audit columns already frozen at the persistence layer (Database Schema §1, §10).
- AC3 — Idempotency where appropriate. Operations that are naturally re-triggerable — every read, and the three Document Generation operations, which the System Architecture (§9) already requires to check for an existing generated output before creating a new one — are specified as idempotent. Operations that record a new historical fact (a Vote, a Discussion Entry, an Approval) are not artificially forced into idempotency, since NFR-02/INV-4 already requires each new fact to be recorded as a new row or version, never deduplicated away.
- AC4 — Explicit validation. Every operation's Validation Rules are stated as a discrete list, each traceable to a named Business Invariant (Domain Model §12) or a Functional/Non-Functional Requirement — never left for the reader to infer from the Outputs alone.
- AC5 — Stable business language. Operation names, DTO field names and error categories use the Ubiquitous Language already frozen in the Domain Model (§2 Bounded Context) and the PRD Glossary (§23) — Decision, Rationale, Approval, Evidence — never storage-shaped nouns such as row, primary key, or table.
- AC6 — No leakage of persistence details. No Data Transfer Object in Section 4 exposes a surrogate key, a version-lineage column, or a table name (Database Schema §1 DP2, §8). Wherever a business identity is required in a request or response, it is named as an opaque Reference (e.g., DecisionRef) whose internal representation is this document's business and no one else's.

2. Service Boundaries

The ten Core Services below are exactly the ten defined in the System Architecture (Phase 1.4, §4); none is renamed, merged, split, or added to. Each row states the public contract surface this document defines for that

service in Section 3 — not its internal implementation, which the System Architecture already governs.

Core Service	Public Contract Scope	Traces To
Authentication & Authorization	Establishes the acting Person/Organization/Role context every other operation requires (AA-01), and authorizes each requested action against it (AA-02, system-internal). No authentication mechanism is defined here, per System Architecture §6.	Architecture §4 Authentication & Authorization
Meeting Management	Creates Meetings, records attendance, and uploads Evidence at Meeting level (MM-01–MM-04), plus the read operations a Client needs to render a Meeting (MM-05–MM-06).	Architecture §4 Meeting Management
Decision Capture	Owns the full Decision lifecycle: proposal, discussion, rationale, alternatives, assumption/risk/obligation/regulation linking, voting and approval (DC-01–DC-10), plus supporting reads (DC-11–DC-12).	Architecture §4 Decision Capture
Decision Graph	Creates and validates Relationship edges between Entities (DG-01, system-internal only — never called directly by the Client Layer, per AP5/AP7), and exposes one bounded, Organization-scoped traversal read (DG-02).	Architecture §4 Decision Graph
Search	Locates candidate Decisions, and secondarily Meetings, Evidence, Assumptions, Risks and Obligations, by keyword or structured filter (SR-01).	Architecture §4 Search
Rationale Retrieval	Compiles one located Decision's full connected rationale (RR-01). Also carries the FR-18 natural-language query contract (RR-02), because the AI Retrieval Layer (Architecture §3, §7) is a read-only interpretive layer over this service, not a Core Service of its own — flagged in Section 9.	Architecture §4 Rationale Retrieval; §7 AI Architecture
Minutes Generation	Compiles a Meeting's Minutes from already-captured data and transitions the Meeting to Concluded (MG-01), plus a read (MG-02).	Architecture §4 Minutes Generation
Resolution Generation	Generates a Resolution from an Approved Decision (RG-01), plus a read (RG-02).	Architecture §4 Resolution Generation
Action Register Generation	Generates an Action Item register from one or more Decisions (ARG-01), plus a read (ARG-02). No status-update operation is defined — see the note below.	Architecture §4 Action Register Generation
Audit & Traceability	A cross-cutting service that decorates every write elsewhere (System Architecture §4) and, at the contract layer, exposes two read-only queries: an audit trail and a version history for any Entity (AT-01–AT-02).	Architecture §4 Audit & Traceability

■ No Notification Service, no eleventh service

Consistent with System Architecture §4's own note, no Notification Service is contracted here — no approved Functional Requirement names one. The AI Retrieval Layer (Architecture §3) is likewise not listed as an eleventh Core Service in Section 2's table; its FR-18 contract is carried by Rationale Retrieval Service (RR-02) because Domain Model §6 states the AI interface “depends entirely on the Rationale Retrieval Service... as its underlying business logic.” This is a judgment call, flagged again in Section 9.

■ No operation to author Regulation

No Core Service below exposes a CreateRegulation-type operation. Domain Model §13 and Database Schema §14 (Item 8) both state Regulation is maintained centrally, by a curation process this phase does not define. DC-08 (Link Regulation) and every other Regulation-referencing operation therefore only reference an already-existing Regulation by RegulationRef; they never create one.

3. Operation Specifications

Every operation below is specified as a business contract — not an HTTP verb, a URL, or a framework annotation. Inputs and Outputs are stated in terms of the Data Transfer Objects defined in Section 4. Validation Rules and Business Errors are stated in terms of the Domain Model Invariants (§12) and PRD Requirements they enforce; the full cross-reference is repeated, in matrix form, in Sections 6 and 9.

3.1 Authentication & Authorization

AA-01 — Establish Actor Context

Field	Detail
Purpose	Resolve an opaque credential/session reference into the Person + Organization + Role context (ActorContext) every other operation requires.
Business Preconditions	The caller holds a credential or session reference issued by a mechanism this document deliberately does not define (System Architecture §6).
Inputs	An opaque CredentialReference (mechanism unspecified).
Outputs	An ActorContext: PersonRef, OrganizationRef, PersonRole, and an issued-at timestamp.
Validation Rules	The credential reference must resolve to exactly one Active Person (Domain Model §11 Person lifecycle); a Departed Person's credential must not resolve to a usable ActorContext.
Business Errors	AuthorizationError (credential does not resolve to an Active Person).
Success Conditions	A valid ActorContext is returned and is usable as the acting-Person parameter of every subsequent operation in the session.
Idempotency Expectations	Idempotent: repeated calls with the same valid credential reference return an equivalent ActorContext.

AA-02 — Authorize Action (system-internal)

Field	Detail
Purpose	Confirm that a given ActorContext's Role is permitted to perform a specific requested operation against a specific target, before that operation's own business logic runs. Invoked by the Application Layer ahead of every Core Service call (System Architecture §3, §6); not called directly by the Client Layer.
Business Preconditions	A valid ActorContext already exists (AA-01).
Inputs	ActorContext; the requested Operation identifier; the target Entity reference (e.g., a MeetingRef or DecisionRef) and its OrganizationRef.
Outputs	A Permit or Deny decision; when Deny, the failing rule reference (Section 5 Authorization Matrix).
Validation Rules	The ActorContext's OrganizationRef must match the target's OrganizationRef (NFR-03, INV-5), except for Regulation, which is Organization-independent shared reference data (Domain Model §13).
Business Errors	AuthorizationError.
Success Conditions	A Permit or Deny decision is returned for every operation invocation; no Core Service operation proceeds without one.
Idempotency Expectations	Idempotent: the same ActorContext, Operation and target always yield the same decision, absent an intervening Role or organizational change.

3.2 Meeting Management

MM-01 — Create Meeting

Field	Detail
Purpose	Create a new Meeting record for an Organization (FR-01).
Business Preconditions	The ActorContext's Role is authorized to author Meetings for the target Organization (Domain Model §13: Corporate Secretary).
Inputs	OrganizationRef; meetingDate; meetingType; an initial list of attending PersonRefs (may be empty and supplemented later via MM-02).
Outputs	A MeetingSummary with status = Open.
Validation Rules	meetingDate and meetingType are required; meetingType must be one of the Domain Model §8 enumerated (illustrative, unconfirmed) values; every attending PersonRef must belong to the target Organization.
Business Errors	ValidationError (missing/malformed field); AuthorizationError.
Success Conditions	A Meeting exists with status = Open and a MeetingCreated Domain Event is raised (Domain Model §7).
Idempotency Expectations	Not idempotent: each call creates a new, distinct Meeting. A caller-supplied idempotency token is out of scope for this phase (flagged, Section 10).

MM-02 — Record Attendance

Field	Detail
Purpose	Add one or more attending Persons to an existing, Open Meeting, satisfying INV-8 before any Decision may be proposed against it.
Business Preconditions	The target Meeting exists and status = Open.
Inputs	MeetingRef; one or more PersonRefs.
Outputs	The updated attendee list on the MeetingSummary.
Validation Rules	Meeting must be Open (Domain Model §11); each PersonRef must belong to the Meeting's Organization; a Person already recorded as attending is not duplicated.
Business Errors	ValidationError; NotFoundError (Meeting or Person); InvariantViolationError (Meeting not Open).
Success Conditions	The Meeting's attendee list includes the named Person(s); INV-8 is satisfiable for any subsequent DC-01 call against this Meeting.
Idempotency Expectations	Idempotent: adding an already-recorded attendee is a no-op, not an error.

MM-03 — Upload Agenda

Field	Detail
Purpose	Attach an agenda document to a Meeting as Evidence (FR-02).
Business Preconditions	The target Meeting exists and status = Open.
Inputs	MeetingRef; a file reference; an optional description.
Outputs	An EvidenceSummary of evidenceType = Agenda, and the SUPPORTED_BY Relationship linking it to the Meeting.
Validation Rules	Meeting must be Open; file reference is required (Domain Model §5 SUPPORTED_BY; Database Schema evidence.file_reference).

Field	Detail
Business Errors	ValidationError; NotFoundError; InvariantViolationError (Meeting not Open).
Success Conditions	The Evidence record exists, organization-scoped (INV-5), and an EvidenceAttached Domain Event is raised.
Idempotency Expectations	Not idempotent: each upload creates a new, distinct Evidence identity (Domain Model §14), even for the same underlying file.

MM-04 — Upload Evidence

Field	Detail
Purpose	Attach a supporting document — financials, a legal opinion, a report — to a Meeting or to a specific Decision (FR-03).
Business Preconditions	The target Meeting exists (and, if Decision-linked, the target Decision exists and belongs to that Meeting).
Inputs	A target reference (MeetingRef or DecisionRef); evidenceType; a file reference; an optional description.
Outputs	An EvidenceSummary and the SUPPORTED_BY Relationship linking it to the named target.
Validation Rules	evidenceType must be one of SupportingDocument, LegalOpinion, FinancialReport, Other (Domain Model §8); if Decision-linked, the Decision must not yet be in a terminal state per INV-4 once Approved/Rejected — later corrections are new Evidence versions, not edits.
Business Errors	ValidationError; NotFoundError; InvariantViolationError (target Decision terminal, per INV-4).
Success Conditions	The Evidence record exists and is retrievable via the target's rationale bundle (RR-01).
Idempotency Expectations	Not idempotent, for the same reason as MM-03.

■ Scope note on MM-04

System Architecture §3 attributes meeting-level Evidence upload to Meeting Management explicitly; Decision-level Evidence upload is inferred to sit in the same service, since FR-03's own wording ("links each to a specific agenda item or Decision") treats upload and linking as one action regardless of target, and Evidence itself is a single Aggregate Root (Domain Model §3) with one upload path. This inference is flagged here rather than silently assumed.

MM-05 — Get Meeting

Field	Detail
Purpose	Retrieve a single Meeting's current state for rendering in the Client Layer.
Business Preconditions	None beyond authorization.
Inputs	MeetingRef.
Outputs	A MeetingSummary, including its attendee list and linked Evidence references.
Validation Rules	MeetingRef must resolve to an existing, current (is_current = true) Meeting version, per Database Schema §8.
Business Errors	NotFoundError; AuthorizationError (Meeting outside the ActorContext's Organization, NFR-03).
Success Conditions	The current MeetingSummary is returned.
Idempotency Expectations	Idempotent by nature (read-only).

MM-06 — List Meetings

Field	Detail
Purpose	Retrieve the set of Meetings belonging to the ActorContext's Organization, to populate a Meeting list view.
Business Preconditions	None beyond authorization.
Inputs	OrganizationRef (implicit from ActorContext); optional filters (date range, status).
Outputs	A list of MeetingSummary (current versions only, by default).
Validation Rules	Results are always scoped to the ActorContext's Organization (NFR-03); never returns another Organization's Meetings.
Business Errors	AuthorizationError.
Success Conditions	The Organization-scoped Meeting list is returned, ordered chronologically (Database Schema §11 indexing recommendation on meeting_date).
Idempotency Expectations	Idempotent by nature (read-only).

■ MM-05 and MM-06 are inferred

Neither is named as a distinct Functional Requirement. Both are included as the minimum read surface the Client Layer needs to render Beachhead Workflow steps 1–5 (PRD §12); this mirrors the flagging pattern Database Schema §1 (DP4) already applied to inferred descriptive fields.

3.3 Decision Capture

DC-01 — Propose Decision

Field	Detail
Purpose	Create a Decision against a Meeting and an agenda item (FR-05).
Business Preconditions	The Meeting exists, status = Open, and has at least one recorded attendee (INV-8).
Inputs	MeetingRef; agendaItemLabel; agendaItemSequence; decisionTitle.
Outputs	A DecisionSummary with status = Proposed.
Validation Rules	INV-1 (exactly one Meeting per Decision, enforced by construction); INV-8 (Meeting must already have ≥1 attendee).
Business Errors	ValidationError; NotFoundError; InvariantViolationError (INV-8 not satisfied).
Success Conditions	The Decision exists with status = Proposed; a DecisionProposed Domain Event is raised; the Decision is MADE_AT-linked to the Meeting.
Idempotency Expectations	Not idempotent: each call proposes a new, distinct Decision.

DC-02 — Record Discussion Entry

Field	Detail
Purpose	Append one point raised during discussion of the agenda item, before the Decision is finalized (FR-04).
Business Preconditions	The Decision exists and has not yet reached a terminal state (Approved/Rejected).
Inputs	DecisionRef; contributorPersonRef; entryText.
Outputs	The appended DiscussionEntry, with its assigned sequenceNumber.
Validation Rules	Decision must not be terminal (INV-4); contributorPersonRef must belong to the Decision's Organization.
Business Errors	ValidationError; NotFoundError; InvariantViolationError (Decision terminal).
Success Conditions	A DiscussionRecorded Domain Event is raised; the Decision's status moves to UnderDiscussion if it was still Proposed (Domain Model §11).
Idempotency Expectations	Not idempotent: each call appends a new, distinct entry.

DC-03 — Record Rationale

Field	Detail
Purpose	Record the reasoning behind a Decision, independent of its eventual outcome (FR-06).
Business Preconditions	The Decision exists and has not yet reached a terminal state.
Inputs	DecisionRef; rationaleText.
Outputs	The updated DecisionSummary with rationaleText populated.
Validation Rules	Decision must not be terminal (INV-4); rationaleText is required before RecordApproval (DC-10) can succeed (INV-2).
Business Errors	ValidationError; NotFoundError; InvariantViolationError (Decision terminal).

Field	Detail
Success Conditions	A RationaleRecorded Domain Event is raised.
Idempotency Expectations	Not strictly idempotent: a later call updates rationaleText for the current (pre-terminal) version rather than versioning a new row, since INV-4 immutability applies only once the Decision reaches Approved/Rejected.

DC-04 — Record Alternative

Field	Detail
Purpose	Record one option considered and not chosen, and why (FR-08).
Business Preconditions	The Decision exists and has not yet reached a terminal state.
Inputs	DecisionRef; descriptionText; reasonNotChosen.
Outputs	The appended Alternative, with its assigned sequenceNumber.
Validation Rules	Decision must not be terminal; both descriptionText and reasonNotChosen are required (Database Schema decision_alternative).
Business Errors	ValidationError; NotFoundError; InvariantViolationError (Decision terminal).
Success Conditions	An AlternativeRecorded Domain Event is raised.
Idempotency Expectations	Not idempotent: each call appends a new, distinct Alternative.

DC-05 — Record Assumption

Field	Detail
Purpose	Record a financial, legal or operational Assumption and link it (BASED_ON) to a Decision (FR-07). May create a new Assumption or reuse an existing one within the same Organization's Decision Graph (INV-5, Success Metric: Decision Graph reuse).
Business Preconditions	The Decision exists and has not yet reached a terminal state.
Inputs	DecisionRef; either assumptionText (to create new) or an existing AssumptionRef (to reuse); optional category.
Outputs	The AssumptionRef (new or reused) and the BASED_ON Relationship.
Validation Rules	If reusing, the AssumptionRef must belong to the same Organization (INV-5) and be non-Superseded (INV-9); Decision must not be terminal.
Business Errors	ValidationError; NotFoundError; AuthorizationError (cross-Organization reuse attempt); InvariantViolationError (Decision terminal, or Superseded Assumption referenced).
Success Conditions	An AssumptionLinked Domain Event is raised; the BASED_ON edge exists (DecisionGraphLinked).
Idempotency Expectations	Idempotent with respect to the edge: linking the same Decision to the same Assumption a second time does not create a duplicate edge.

DC-06 — Link Risk

Field	Detail
Purpose	Record or reuse a Risk knowingly accepted as part of a Decision, and link it (ACCEPTS) (FR-09).
Business Preconditions	The Decision exists and has not yet reached a terminal state.

Field	Detail
Inputs	DecisionRef; either riskText (to create new) or an existing RiskRef (to reuse).
Outputs	The RiskRef and the ACCEPTS Relationship.
Validation Rules	Same Organization-scoping and non-Superseded rules as DC-05 (INV-5, INV-9); no severity/likelihood fields are accepted — PRD §18 places a standalone Risk Register out of scope, and this schema-level exclusion (Database Schema §3, FLAG) carries through to the contract.
Business Errors	ValidationError; NotFoundError; AuthorizationError; InvariantViolationError.
Success Conditions	A RiskAccepted Domain Event is raised; the ACCEPTS edge exists.
Idempotency Expectations	Idempotent with respect to the edge, as DC-05.

DC-07 — Link Obligation

Field	Detail
Purpose	Record or reuse a compliance/contractual Obligation and link it (SUBJECT_TO) to a Decision (FR-09).
Business Preconditions	The Decision exists and has not yet reached a terminal state.
Inputs	DecisionRef; either obligationText plus a RegulationRef (to create new) or an existing ObligationRef (to reuse).
Outputs	The ObligationRef and the SUBJECT_TO Relationship.
Validation Rules	A new Obligation requires a RegulationRef per the Domain Model's DERIVED_FROM many:1 cardinality, carried through from Database Schema §14 (Item 4) — this is an inherited tension the Schema itself flagged (some Obligations are contractual, not regulation-derived) and is not re-resolved here; Organization-scoping and non-Superseded rules as DC-05/DC-06.
Business Errors	ValidationError; NotFoundError; AuthorizationError; InvariantViolationError.
Success Conditions	An ObligationLinked Domain Event is raised; the SUBJECT_TO edge exists.
Idempotency Expectations	Idempotent with respect to the edge, as DC-05.

DC-08 — Link Regulation

Field	Detail
Purpose	Link a Decision to the Regulation it was reasoned against (GOVERNED_BY).
Business Preconditions	The Decision exists and has not yet reached a terminal state; the target Regulation already exists (Regulation is not authored through this API — Section 2 note).
Inputs	DecisionRef; RegulationRef.
Outputs	The GOVERNED_BY Relationship.
Validation Rules	RegulationRef must resolve to an existing, non-Superseded Regulation (INV-9); Regulation is shared reference data and is not Organization-scoped (Domain Model §13), so no cross-Organization check applies here, unlike DC-05–DC-07.
Business Errors	ValidationError; NotFoundError; InvariantViolationError (Decision terminal, or Superseded Regulation).
Success Conditions	The GOVERNED_BY edge exists (DecisionGraphLinked).
Idempotency Expectations	Idempotent with respect to the edge, as DC-05.

■ Scope note on DC-08

PRD FR-09's own text names only Risk & Obligation Linking, but Domain Model §10 explicitly traces the GOVERNED_BY Relationship Type to FR-09 as well ("A Decision was made with a specific Regulation in view... Traces to: FR-09"). This operation follows the Domain Model's own trace rather than the PRD's narrower prose, consistent with the Domain Model being the frozen, more specific artifact on this point.

DC-09 — Cast Vote

Field	Detail
Purpose	Record a Person's For / Against / Abstain choice on a Decision (FR-10).
Business Preconditions	The Decision exists; the voting Person has not already voted on this Decision (INV-6).
Inputs	DecisionRef; voterPersonRef; voteChoice (For Against Abstain).
Outputs	The recorded Vote, and the updated tally visible on the DecisionSummary.
Validation Rules	voteChoice must be one of the three enumerated values (Domain Model §8); INV-6 — at most one Vote per Person per Decision.
Business Errors	ValidationError; NotFoundError; ConflictError (a Vote already exists for this Person and Decision, INV-6).
Success Conditions	A VoteCast Domain Event is raised; the Decision's status moves to Voted once voting is recorded (Domain Model §11).
Idempotency Expectations	Not idempotent — a second call for the same Person/Decision pair is a ConflictError, not a silent overwrite, per INV-6.

DC-10 — Record Approval

Field	Detail
Purpose	Formally mark a Decision as Approved (or Rejected), capturing the approving Person(s) and the approval date (FR-11).
Business Preconditions	The Decision has recorded Rationale, at least one Vote, and at least one Approval-in-progress (INV-2, enforced by the Decision Approval Eligibility Service, Domain Model §6).
Inputs	DecisionRef; approverPersonRef; outcome (Approved Rejected).
Outputs	The finalized DecisionSummary with status = Approved or Rejected.
Validation Rules	INV-2 — Rationale present, ≥1 Vote, ≥1 Approval, all checked before the transition is permitted.
Business Errors	ValidationError; NotFoundError; InvariantViolationError (INV-2 not satisfied); ConflictError (Decision already terminal).
Success Conditions	A DecisionApproved or DecisionRejected Domain Event is raised; per INV-4, the Decision's Rationale, Assumptions, Alternatives, Risk/Obligation links, Votes and Approvals become immutable from this point — any later correction is a new version, never an edit.
Idempotency Expectations	Not idempotent: a second call against an already-terminal Decision is a ConflictError, consistent with INV-4.

DC-11 — Get Decision

Field	Detail
Purpose	Retrieve a single Decision's current state, including its embedded Value Objects (discussion, alternatives, votes, approvals).

Field	Detail
Business Preconditions	None beyond authorization.
Inputs	DecisionRef.
Outputs	A DecisionSummary with embedded DiscussionEntry, Alternative, Vote and Approval collections.
Validation Rules	DecisionRef must resolve to an existing, current version (Database Schema §8).
Business Errors	NotFoundError; AuthorizationError.
Success Conditions	The current DecisionSummary is returned.
Idempotency Expectations	Idempotent by nature (read-only).

DC-12 — List Decisions For Meeting

Field	Detail
Purpose	Retrieve every Decision proposed against a given Meeting, to populate a Meeting's Decision list view.
Business Preconditions	None beyond authorization.
Inputs	MeetingRef.
Outputs	A list of DecisionSummary (current versions only).
Validation Rules	Results scoped to the Meeting's Organization (NFR-03).
Business Errors	NotFoundError; AuthorizationError.
Success Conditions	The Decision list for the Meeting is returned.
Idempotency Expectations	Idempotent by nature (read-only).

■ DC-11 and DC-12 are inferred

As with MM-05/MM-06, neither is a distinct Functional Requirement; both are the minimum read surface Beachhead Workflow steps 4–10 (PRD §12) require the Client Layer to have.

3.4 Decision Graph

DG-01 — Create Relationship (system-internal)

Field	Detail
Purpose	Create and validate a Relationship (graph edge) between two Entities (FR-15). Invoked by other Core Services immediately after each of their own writes (e.g., DC-05 calling this after recording an Assumption) — never called directly by the Client Layer, per AP5/AP7.
Business Preconditions	Both endpoint Entities already exist and are not Superseded (INV-9).
Inputs	fromEntityRef; toEntityRef; relationshipType (one of the fourteen named types, Domain Model §9); any edge-carried attribute (e.g., VoteChoice for CAST_VOTE_ON — though DC-09 carries this directly, per Database Schema §3's associative-table treatment).
Outputs	The created Relationship, and a DecisionGraphLinked Domain Event.
Validation Rules	INV-9 — both ends exist, are non-Superseded, and the (fromType, toType, relationshipType) triple is one of the fourteen permitted combinations (Domain Model §9); INV-5 for the four Organization-scoped reusable Entities (Evidence, Assumption, Risk, Obligation).
Business Errors	NotFoundError; InvariantViolationError (INV-9 or INV-5 violation).
Success Conditions	The edge is persisted and immediately traversable via DG-02 and RR-01.
Idempotency Expectations	Idempotent: creating an already-existing edge (same from, to, type) is a no-op, not a duplicate.

DG-02 — Get Entity Relationships

Field	Detail
Purpose	Retrieve the directly connected Relationship edges for a given Entity, supporting FR-15's description of a “queryable graph” independent of the bounded rationale fan-out RR-01 already provides for one Decision.
Business Preconditions	None beyond authorization.
Inputs	An EntityRef of any of the eleven Entity types; an optional relationshipType filter.
Outputs	A list of Relationship edges, each naming its type, direction and the connected EntityRef.
Validation Rules	Results exclude edges to Superseded or cross-Organization Entities (INV-9, INV-5), except where Regulation is the connected Entity.
Business Errors	NotFoundError; AuthorizationError.
Success Conditions	The Entity's current, permitted edge set is returned.
Idempotency Expectations	Idempotent by nature (read-only).

■ DG-02 is inferred

FR-15 describes the graph as queryable in the abstract but does not name a distinct retrieval operation beyond FR-16 (which RR-01/SR-01 already implement for Decision rationale specifically). DG-02 is included as the minimal, generic complement — flagged rather than silently assumed.

3.5 Search

SR-01 — Search Decisions

Field	Detail
Purpose	Locate candidate Decisions — and secondarily Meetings, Evidence, Assumptions, Risks and Obligations — matching a keyword or structured filter (FR-16, search half).
Business Preconditions	None beyond authorization.
Inputs	A keyword string or structured filter (date range, status, PersonRef); OrganizationRef (implicit from ActorContext).
Outputs	A ranked or matching list of EntityRefs with a short display label each.
Validation Rules	Results are always scoped to the ActorContext's Organization (NFR-03); operates over the derived, text-searchable read model (System Architecture AP6, Database Schema §11), never the canonical tables directly.
Business Errors	ValidationError (malformed filter); AuthorizationError.
Success Conditions	A candidate set is returned for the caller to narrow via RR-01.
Idempotency Expectations	Idempotent by nature (read-only).

3.6 Rationale Retrieval

RR-01 — Retrieve Decision Rationale

Field	Detail
Purpose	Given one located Decision, compile its full connected rationale — Discussion, Assumptions, Alternatives, Risks, Obligations, Evidence, Regulations, Votes, Approvals (FR-16, retrieve half).
Business Preconditions	The target Decision exists.
Inputs	DecisionRef.
Outputs	A RationaleBundle (Section 4).
Validation Rules	Fan-out is the fixed, bounded join list defined in Database Schema §12 — not an open-ended traversal; results scoped to the ActorContext's Organization (NFR-03).
Business Errors	NotFoundError; AuthorizationError.
Success Conditions	The complete RationaleBundle for the current version of the Decision is returned, each element attributed to its source record (traceability, NFR-01).
Idempotency Expectations	Idempotent by nature (read-only).

RR-02 — Query Decision Memory (Natural Language)

Field	Detail
Purpose	Accept a natural-language question and return an answer grounded strictly in already-captured Decision Graph rationale (FR-18).
Business Preconditions	None beyond authorization; the query is constrained to the ActorContext's Organization before any retrieval occurs (System Architecture §7).
Inputs	A natural-language queryText; ActorContext (implicit).
Outputs	A synthesized answer, plus the list of source EntityRefs (Decision, Discussion Entry, Assumption, Risk, Obligation, Evidence) each claim in the answer is attributed to.

Field	Detail
Validation Rules	The response must not include any claim that cannot be traced to a specific record returned by RR-01's fan-out for the relevant Decision(s) (System Architecture §7 Hallucination Mitigation); where no rationale is found, the response must say so rather than infer one.
Business Errors	ValidationError (empty query); AuthorizationError.
Success Conditions	An answer is returned with full source attribution, or an explicit "no recorded rationale" response — never a fabricated one.
Idempotency Expectations	Idempotent with respect to underlying data: the same query against an unchanged Decision Graph returns an equivalent, equally-grounded answer. This operation performs no write of its own back into the Decision Graph (System Architecture AP5).

■ RR-02's service placement is a judgment call

FR-18 is explicitly "an application/interface-layer concern... not a Domain Service itself" (Domain Model §6), realized architecturally as the AI Retrieval Layer (System Architecture §3, §7) sitting over Search and Rationale Retrieval. Because Domain Model §6 names Rationale Retrieval Service as its sole underlying business logic dependency, this document carries RR-02's contract under Rationale Retrieval Service rather than inventing an eleventh Core Service — flagged here and in Section 2, consistent with the instruction not to invent additional services.

3.7 Minutes Generation

MG-01 — Generate Minutes

Field	Detail
Purpose	Compile a Meeting's Discussion, Decisions, Votes and Approvals into the Minutes Value Object, and conclude the Meeting (FR-12).
Business Preconditions	The Meeting exists and status = Open; no Minutes have already been generated for it (System Architecture §9 duplicate-generation check).
Inputs	MeetingRef.
Outputs	A MinutesDocument.
Validation Rules	No new information may be entered as part of this call — the output is compiled entirely from data already captured (PRD §21 “no re-entry of information” acceptance criterion); Meeting must not already have generated Minutes.
Business Errors	NotFoundError; InvariantViolationError (Meeting not Open); ConflictError (Minutes already generated).
Success Conditions	The Minutes exist; the Meeting transitions Open → Concluded; a MinutesGenerated Domain Event is raised.
Idempotency Expectations	Idempotent by construction: a retried call against a Meeting that already has generated Minutes returns the existing MinutesDocument rather than creating a duplicate (System Architecture §9).

MG-02 — Get Minutes

Field	Detail
Purpose	Retrieve a Meeting's generated Minutes.
Business Preconditions	Minutes have already been generated (MG-01).
Inputs	MeetingRef.
Outputs	The MinutesDocument, if one exists.
Validation Rules	None beyond existence and Organization scoping.
Business Errors	NotFoundError (no Minutes yet generated); AuthorizationError.
Success Conditions	The MinutesDocument is returned.
Idempotency Expectations	Idempotent by nature (read-only).

3.8 Resolution Generation

RG-01 — Generate Resolution

Field	Detail
Purpose	Generate a Resolution document from an Approved Decision (FR-13).
Business Preconditions	The Decision's status = Approved (INV-3); no Resolution has already been generated for it.
Inputs	DecisionRef.
Outputs	A ResolutionDocument, including its durable resolutionReference citation.
Validation Rules	INV-3 — generation permitted only when decision.status = Approved.

Field	Detail
Business Errors	NotFoundError; InvariantViolationError (INV-3 not satisfied, e.g., Decision is Rejected or not yet terminal); ConflictError (Resolution already generated).
Success Conditions	The Resolution exists, is immutable from generation onward (Domain Model §11), and RESULTS_IN-linked to the Decision; a ResolutionGenerated Domain Event is raised.
Idempotency Expectations	Idempotent by construction, as MG-01: a retried call against a Decision that already has a generated Resolution returns the existing ResolutionDocument.

RG-02 — Get Resolution

Field	Detail
Purpose	Retrieve a Decision's generated Resolution.
Business Preconditions	A Resolution has already been generated (RG-01).
Inputs	DecisionRef or ResolutionRef.
Outputs	The ResolutionDocument, if one exists.
Validation Rules	None beyond existence and Organization scoping.
Business Errors	NotFoundError; AuthorizationError.
Success Conditions	The ResolutionDocument is returned.
Idempotency Expectations	Idempotent by nature (read-only).

3.9 Action Register Generation

ARG-01 — Generate Action Register

Field	Detail
Purpose	Generate Action Item entries from one or more Decisions within a Meeting that require follow-up (FR-14).
Business Preconditions	Each named Decision exists; INV-7 is satisfiable (each Action Item will resolve to exactly one Decision and exactly one owning Person).
Inputs	One or more DecisionRefs requiring follow-up; for each, an assignedToPersonRef and the inferred descriptive fields noted below.
Outputs	A list of ActionItem records (the Action Register).
Validation Rules	INV-7 — each Action Item RESULTS_IN from exactly one Decision and is ASSIGNED_TO exactly one Person; assignedToPersonRef must belong to the Decision's Organization.
Business Errors	ValidationError; NotFoundError; InvariantViolationError (INV-7 not satisfied).
Success Conditions	Each Action Item exists, RESULTS_IN- and ASSIGNED_TO-linked; an ActionItemGenerated Domain Event is raised per item.
Idempotency Expectations	Idempotent by construction with respect to a given Decision/assignee pairing already generated, mirroring MG-01/RG-01's duplicate-generation guard (System Architecture §9), though PRD/Domain Model do not state this explicitly for Action Items — flagged as an extension of the same principle applied uniformly (Section 10).

ARG-02 — Get Action Register

Field	Detail
Purpose	Retrieve the generated Action Items for a Meeting or a Decision.
Business Preconditions	None beyond authorization.
Inputs	MeetingRef or DecisionRef.
Outputs	A list of ActionItem records.
Validation Rules	None beyond existence and Organization scoping.
Business Errors	NotFoundError; AuthorizationError.
Success Conditions	The current Action Register for the named target is returned.
Idempotency Expectations	Idempotent by nature (read-only).

■ No status-update operation for Action Items

Domain Model §8 explicitly defers an ActionItemStatus enum to Future Expansion, and Database Schema §3 mirrors this as a deliberate omission. No UpdateActionItemStatus (or similar) operation is defined here for the same reason — adding one would model a lifecycle not approved for V1.

3.10 Audit & Traceability

AT-01 — Get Audit Trail

Field	Detail
Purpose	Retrieve the acting-Person and timestamp of record for any Entity, directly serving the Auditor persona's stated need ("Show us why this decision was approved," PRD §10).
Business Preconditions	The target Entity exists.
Inputs	An EntityRef of any of the eleven Entity types.
Outputs	createdAt; createdByPersonRef; and, where applicable, the same for each linked Relationship the fan-out includes (mirroring RR-01's bounded join list).
Validation Rules	None beyond existence and Organization scoping (NFR-03).
Business Errors	NotFoundError; AuthorizationError.
Success Conditions	The complete NFR-01 audit record for the Entity is returned.
Idempotency Expectations	Idempotent by nature (read-only).

AT-02 — Get Version History

Field	Detail
Purpose	Retrieve the full version lineage for a versioned Entity (Meeting, Decision, Evidence, Assumption, Risk, Obligation, Regulation — the seven tables Database Schema §8 marks Versioned: Yes), supporting a "what did we believe when we approved this" reconstruction (Database Schema §8).
Business Preconditions	The target Entity is one of the seven versioned types.
Inputs	An EntityRef.
Outputs	An ordered list of versions, each with versionNumber, createdAt, createdByPersonRef, and isCurrent.
Validation Rules	Only defined for the seven versioned Entity types; Resolution, Action Item and Minutes have no version history to return, per Database Schema §8's explicit exclusion.
Business Errors	NotFoundError; ValidationError (Entity type has no version lineage).
Success Conditions	The full supersedes_id chain, expressed in business terms (no internal key names), is returned in order.
Idempotency Expectations	Idempotent by nature (read-only).

■ AT-01 and AT-02 are inferred

Neither is a distinct Functional Requirement; both operationalize NFR-01/NFR-02's traceability guarantee as a queryable contract, since a guarantee an Auditor cannot query is not, in practice, auditable.

4. Data Transfer Objects (Logical)

Every shape below is defined in business terms only. No field names a surrogate key, a version-lineage column, or a table (AC6). Reference types (suffixed Ref) are opaque business identifiers; their internal representation is not part of this contract.

4.1 Reference Types

DTO	Represents
OrganizationRef	A single Organization (the tenancy boundary, NFR-03).
PersonRef	A single Person, stable across the Active → Departed transition (Domain Model §11, §14).
MeetingRef	A single Meeting occurrence.
DecisionRef	A single Decision, stable across its full lifecycle including versioned corrections.
EvidenceRef / AssumptionRef / RiskRef / ObligationRef	A single reusable reference Entity, scoped to one Organization's Decision Graph (INV-5).
RegulationRef	A single Regulation citation, Organization-independent (Domain Model §13).
ResolutionRef / ActionItemRef	A single Entity, child of exactly one Decision (Domain Model §4).
EntityRef	A discriminated union of any of the above eleven Entity reference types, used where an operation is generic across Entity kinds (e.g., DG-02, AT-01, AT-02).

4.2 Context and Actor

DTO	Fields
ActorContext	personRef; organizationRef; role (PersonRole); issuedAt.

4.3 Meeting and Decision Shapes

DTO	Fields
MeetingSummary	meetingRef; organizationRef; meetingDate; meetingType; status (Open Concluded); attendees: PersonRef[]; linkedEvidence: EvidenceRef[]; concludedAt?.
DecisionSummary	decisionRef; meetingRef; agendaItemLabel; agendaItemSequence; decisionTitle; status (Proposed UnderDiscussion Voted Approved Rejected); rationaleText?; discussion: DiscussionEntry[]; alternatives: Alternative[]; votes: Vote[]; approvals: Approval[]; createdAt; createdByPersonRef.
DiscussionEntry	contributorPersonRef; entryText; sequenceNumber; recordedAt.
Alternative	descriptionText; reasonNotChosen; sequenceNumber.
Vote	voterPersonRef; voteChoice (For Against Abstain); castAt.
Approval	approverPersonRef; approvedAt.

4.4 Reusable Reference Entity Summaries

DTO	Fields
EvidenceSummary	evidenceRef; evidenceType (Agenda SupportingDocument LegalOpinion FinancialReport Other); fileReference; description?; uploadedByPersonRef; uploadedAt.
AssumptionSummary / RiskSummary / ObligationSummary	ref; text; (Obligation only) linkedRegulationRef; category? (Assumption only, inferred, Database Schema §14 Item 2); recordedByPersonRef; createdAt.
RegulationSummary	regulationRef; citation; title; jurisdiction?; sourceReference?.

4.5 Compiled Outputs

DTO	Fields
RationaleBundle	decision: DecisionSummary; discussion: DiscussionEntry[]; alternatives: Alternative[]; votes: Vote[]; approvals: Approval[]; assumptions: AssumptionSummary[]; risks: RiskSummary[]; obligations: ObligationSummary[]; evidence: EvidenceSummary[]; regulations: RegulationSummary[].
MinutesDocument	minutesRef; meetingRef; generatedAt; compiledContent.
ResolutionDocument	resolutionRef; decisionRef; resolutionReference; generatedAt; contentReference.
ActionItem	actionItemRef; decisionRef; assignedToPersonRef; descriptionText (inferred, Database Schema §14 Item 1); dueDate? (inferred, Database Schema §14 Item 1); generatedAt.

4.6 Audit Shapes

DTO	Fields
AuditRecord	entityRef; createdAt; createdByPersonRef; linkedRecordAudits: AuditRecord[] (bounded to RR-01's fan-out).
VersionHistoryEntry	versionNumber; createdAt; createdByPersonRef; isCurrent.

5. Authorization Matrix

Roles below are exactly the PersonRole enumeration frozen in the Domain Model (§8): Director, CorporateSecretary, ExternalCounsel, Auditor, Other. “System” denotes an operation whose output is authored by the system of record itself, not by an individual Person (Domain Model §13); a human Role may still be the one requesting that the System act. This matrix is consistent with, and adds no permission beyond, Domain Model §13’s Ownership Boundaries and System Architecture §6’s Authorization section.

Operation	Permitted Roles	Basis
AA-01 Establish Actor Context	Any credential holder (pre-authorization step)	System Architecture §6
AA-02 Authorize Action	System-internal only	System Architecture §3, §6
MM-01 Create Meeting	CorporateSecretary	Domain Model §13 (Meeting owned by Corporate Secretary)
MM-02 Record Attendance	CorporateSecretary	Domain Model §13
MM-03 Upload Agenda	CorporateSecretary	Domain Model §13 (Meeting)
MM-04 Upload Evidence	CorporateSecretary; ExternalCounsel	Domain Model §13 (Evidence — whoever uploads it)
MM-05 Get Meeting / MM-06 List Meetings	Director; CorporateSecretary; ExternalCounsel; Auditor; Other	Organization-scoped read (NFR-03); no PRD ownership restriction on reads
DC-01 Propose Decision	CorporateSecretary	Domain Model §13 (Corporate Secretary records the Decision)
DC-02 Record Discussion Entry / DC-03 Record Rationale / DC-04 Record Alternative	CorporateSecretary; Director; ExternalCounsel	Domain Model §13 (“collaboratively authored”)
DC-05 Record Assumption / DC-06 Link Risk / DC-07 Link Obligation / DC-08 Link Regulation	CorporateSecretary; Director; ExternalCounsel	Domain Model §13 (“whoever raises them during discussion or decision capture”)
DC-09 Cast Vote	Director	PRD §10 Persona (“Board Director / CEO ... votes, approves”); Domain Model §13
DC-10 Record Approval	Director	PRD §10 Persona; Domain Model §13
DC-11 Get Decision / DC-12 List Decisions For Meeting	Director; CorporateSecretary; ExternalCounsel; Auditor; Other	Organization-scoped read
DG-01 Create Relationship	System-internal only	System Architecture AP5, AP7
DG-02 Get Entity Relationships	Director; CorporateSecretary; ExternalCounsel; Auditor; Other	Organization-scoped read; FR-15 “queryable graph”
SR-01 Search Decisions	Director; CorporateSecretary; ExternalCounsel; Auditor; Other	FR-16 (“User can search...”, generic)
RR-01 Retrieve Decision Rationale / RR-02 Query Decision Memory	Director; CorporateSecretary; ExternalCounsel; Auditor; Other	FR-16, FR-18; PRD §10 Auditor persona named explicitly
MG-01 Generate Minutes / RG-01 Generate Resolution / ARG-01 Generate Action Register	Requestable by CorporateSecretary; authored by System	Domain Model §13 (“Generated by the relevant Domain Service... not manually authored by any actor”)

Operation	Permitted Roles	Basis
MG-02 Get Minutes / RG-02 Get Resolution / ARG-02 Get Action Register	Director; CorporateSecretary; ExternalCounsel; Auditor; Other	Organization-scoped read
AT-01 Get Audit Trail / AT-02 Get Version History	Director; CorporateSecretary; ExternalCounsel; Auditor; Other	NFR-01, NFR-02; PRD §10 Auditor persona named explicitly

■ No Regulation-authoring role

No row above grants any Role the ability to create or edit a Regulation, consistent with Section 2's note: Regulation is centrally curated (Domain Model §13), outside this API's write surface entirely.

6. Validation & Business Rules

Every Business Invariant frozen in the Domain Model (§12) is enforced by at least one operation above. The table below is the inverse index: for each Invariant, which operation(s) enforce it, and at what point in the contract.

Invariant	Rule (abbreviated)	Enforced By	Enforcement Point
INV-1	A Decision is MADE_AT exactly one Meeting.	DC-01	Validation Rule (structural — a Decision cannot be proposed without a MeetingRef)
INV-2	A Decision cannot become Approved without Rationale, ≥ 1 Vote, ≥ 1 Approval.	DC-10	Business Error (InvariantViolationError) at the moment RecordApproval is requested
INV-3	A Resolution can only be generated for an Approved Decision.	RG-01	Business Error (InvariantViolationError)
INV-4	Post-terminal Decision content is immutable; corrections are new versions.	DC-02–DC-10 (write guard); AT-02 (read of the resulting lineage)	Business Error (InvariantViolationError / ConflictError) on any write attempt against a terminal Decision
INV-5	Evidence/Assumption/Risk/Obligation reuse stays within one Organization.	DC-05–DC-07, DG-01	Business Error (AuthorizationError / InvariantViolationError)
INV-6	At most one Vote per Person per Decision.	DC-09	Business Error (ConflictError)
INV-7	An Action Item has exactly one Decision and exactly one owning Person.	ARG-01	Validation Rule / Business Error (InvariantViolationError)
INV-8	A Meeting needs ≥ 1 attendee before a Decision can be created against it.	MM-02 (satisfies); DC-01 (checks)	Business Error (InvariantViolationError) at Decision proposal
INV-9	Every Relationship connects two existing, non-superseded, permitted-type Entities.	DG-01 (all linking operations route through it)	Business Error (NotFoundError / InvariantViolationError)
INV-10	No permanent deletion of Approved-linked records.	All operations (no Delete operation is defined anywhere in Section 3)	Structural — absence of a delete contract, not a runtime check

■ NFR mapping

NFR-01 (traceability) and NFR-02 (versioning) are not single-operation concerns; they are enforced uniformly by AC2 (Section 1) and by every write operation's implicit ActorContext/timestamp attribution, per AT-01/AT-02's read contract over that same data. NFR-03 (confidentiality) is enforced by AA-02 ahead of every operation, not listed per-row above to avoid repeating the same line thirty-three times.

7. Error Contract

Five logical error categories cover every Business Error named in Section 3. No transport-specific status code is prescribed; each category is a business-meaning distinction, not a wire-format one.

Category	Raised When	Distinguishing Characteristic	Example Operations
ValidationError	A request's input fails a stated Validation Rule before any business logic runs — a missing required field, a malformed enum value, an empty query.	Detectable from the request alone, independent of the current state of the Decision Graph.	MM-01, DC-01, DC-09, SR-01
AuthorizationError	The ActorContext's Role is not permitted to perform the requested operation (Section 5), or the target Entity is outside the Actor's Organization scope (NFR-03, INV-5).	Depends on who is asking, not on what is being asked.	AA-02, MM-01, DC-05, DG-01
NotFoundError	A referenced Entity does not exist, or exists only as a Superseded / non-current version and the operation requires the current one.	The reference itself is well-formed but does not resolve.	MM-05, DC-11, RG-02, AT-02
InvariantViolationError	The request is well-formed and authorized, but performing it would violate a named Business Invariant given the current state of the Decision Graph.	Depends on the current state of the graph, not on the request's shape or the caller's identity.	DC-10 (INV-2), RG-01 (INV-3), ARG-01 (INV-7)
ConflictError	The target's state has changed since the caller last read it, or the request violates a business-level uniqueness constraint.	A second, otherwise-valid attempt at an already-settled fact.	DC-09 (INV-6), DC-10 (re-approval), MG-01 / RG-01 (already generated)

■ Every error is attributable

Consistent with AC2, every error response carries the ActorContext and timestamp of the failed attempt, so that a failed request is itself part of the traceable record — not merely a discarded one.

8. Versioning Strategy

This contract must remain at least as durable as the persistence layer it fronts (Database Schema §8–§9). The strategy below applies the same no-overwrite discipline to the contract itself.

- Additive-first evolution. New optional fields and new operations may be added to a Core Service's contract without breaking existing callers; a caller that does not send or expect the new field continues to function unchanged.
- An explicit contract version accompanies every request and response at the logical level — not tied to any URL or transport — letting the backend detect and reject a call made against a retired contract shape, mirroring the versionNumber pattern already frozen for data (Database Schema §8).
- Deprecation, not deletion. A retired operation or field is marked Deprecated and continues to function through a governance-approved transition window before removal — the contract layer should not be less durable than the no-hard-delete persistence layer it fronts (Database Schema §9, INV-10).
- Breaking changes — removing a field, changing what a Validation Rule means, renaming a business concept — require the same founder/governance approval as any other scope change (PRD §19 Product Constraints), since a contract change of that kind is effectively a reinterpretation of an approved artifact, which the Decision Freeze prohibits without explicit sign-off.
- Every new operation or field must trace to an already-approved requirement (Section 9). An operation that cannot be traced is out of scope for this phase, exactly as AP8 (System Architecture) and DP4 (Database Schema) already established for their respective layers.

9. Traceability Matrix

Every operation defined in Section 3 is mapped below to the specific point in each frozen upstream artifact that justifies its existence. Cells marked — indicate no direct object-level mapping in that artifact for this specific operation, stated explicitly rather than left blank without explanation.

Operation	Product Constitution	PRD	Domain Model	Database Schema	System Architecture
AA-01 Establish Actor Context	—	NFR-03	§13 Ownership Boundaries	—	§4, §6
AA-02 Authorize Action	—	NFR-03	§13	—	§3, §6
MM-01 Create Meeting	Principle: capture reasoning at decision time	FR-01	§3 Meeting Aggregate	meeting table	§4 Meeting Management
MM-02 Record Attendance	—	FR-01	INV-8	meeting_attendee	§4; Flow 1
MM-03 Upload Agenda	Philosophy: “Documents are evidence”	FR-02	§5 SUPPORTED_BY	evidence table	§4; Flow 1
MM-04 Upload Evidence	Philosophy: “Documents are evidence”	FR-03	§5, §9–10	evidence table	§4; Flow 1
MM-05 Get Meeting	—	— (inferred)	—	meeting table	§3 Client Layer
MM-06 List Meetings	—	— (inferred)	—	meeting table	§3 Client Layer
DC-01 Propose Decision	Philosophy: “Decisions are knowledge”	FR-05	§3 Decision Aggregate; INV-1, INV-8	decision table	§4 Decision Capture; Flow 2
DC-02 Record Discussion Entry	—	FR-04	§5 Discussion Entry	decision_discussion_entry	Flow 2
DC-03 Record Rationale	Philosophy: “Decisions are knowledge”	FR-06	§6 Decision Approval Eligibility Service	decision.rationale_text	Flow 2
DC-04 Record Alternative	Category Thesis: “alternatives considered”	FR-08	§5 Alternative	decision_alternative	Flow 2
DC-05 Record Assumption	—	FR-07	§3 Assumption Aggregate	assumption table; decision_assumption_link	Flow 2
DC-06 Link Risk	—	FR-09	§3 Risk Aggregate	risk table; decision_risk_link	Flow 2
DC-07 Link Obligation	—	FR-09	§3 Obligation Aggregate	obligation table; decision_obligation_link	Flow 2
DC-08 Link Regulation	—	FR-09 (per Domain Model trace)	§10 GOVERNED_BY	decision_regulation_link	Flow 2
DC-09 Cast Vote	—	FR-10	§5 Vote; INV-6	decision_vote	Flow 3

Operation	Product Constitution	PRD	Domain Model	Database Schema	System Architecture
DC-10 Record Approval	Philosophy: "Decisions are knowledge"	FR-11	§5 Approval; INV-2, INV-4	decision_approval	Flow 3
DC-11 Get Decision	—	— (inferred)	—	decision table	§3 Client Layer
DC-12 List Decisions For Meeting	—	— (inferred)	—	decision table	§3 Client Layer
DG-01 Create Relationship	Philosophy: "the memory graph is the product"	FR-15	§6 Decision Graph Linking Service; INV-9	§4–6 (Relationship Layer)	§4 Decision Graph
DG-02 Get Entity Relationships	Philosophy: "the memory graph is the product"	FR-15 ("queryable graph")	§9–10	— (derived read)	§4 Decision Graph
SR-01 Search Decisions	North Star: reusable, searchable rationale	FR-16 (search half)	— (architectural refinement)	§11 indexing	§4 Search
RR-01 Retrieve Decision Rationale	North Star: reusable, searchable rationale	FR-16 (retrieve half)	§6 Rationale Retrieval Service	§12 bounded fan-out	§4 Rationale Retrieval
RR-02 Query Decision Memory	Philosophy: "AI is the interface"	FR-18	§6 (AI Retrieval Interface note)	§11–12	§7 AI Architecture
MG-01 Generate Minutes	—	FR-12	§6 Minutes Compilation Service	minutes table	§4 Minutes Generation; Flow 4
MG-02 Get Minutes	—	— (inferred)	§5 Minutes VO	minutes table	§3 Client Layer
RG-01 Generate Resolution	—	FR-13	§6 Resolution Drafting Service; INV-3	resolution table	§4 Resolution Generation; Flow 4
RG-02 Get Resolution	—	— (inferred)	—	resolution table	§3 Client Layer
ARG-01 Generate Action Register	—	FR-14	§6 Action Register Compilation Service; INV-7	action_item table	§4 Action Register Generation; Flow 4
ARG-02 Get Action Register	—	— (inferred)	—	action_item table	§3 Client Layer
AT-01 Get Audit Trail	Principle: preserve traceability	NFR-01	—	§1, §10	§4 Audit & Traceability
AT-02 Get Version History	Principle: preserve traceability	NFR-02	INV-4	§8 version lineage	§4 Audit & Traceability

10. Deferred Items and Flagged Inferences

Consistent with the flagging discipline of Phases 1.1–1.4, the items below are surfaced explicitly rather than silently decided. None represents a new business capability; each is either a minimal necessary contract detail or an open question carried forward for governance.

#	Item	Treatment in This Document
1	Read operations (MM-05/06, DC-11/12, MG-02, RG-02, ARG-02, DG-02, AT-01/02).	Included as the minimal Client-facing read surface the Beachhead Workflow requires; none is a distinct Functional Requirement, and each is flagged inline in Section 3 where introduced.
2	RR-02's placement under Rationale Retrieval Service rather than a distinct AI Retrieval Service.	A judgment call, made because Domain Model §6 names Rationale Retrieval Service as FR-18's sole underlying business-logic dependency; flagged in Sections 2 and 3.
3	MM-04's ownership of Decision-linked Evidence upload.	Inferred from FR-03's combined upload-and-link wording and Evidence's single-Aggregate-Root shape; flagged in Section 3.
4	DC-08 (Link Regulation) traced to FR-09 via the Domain Model rather than the PRD's narrower prose.	Follows the Domain Model's own explicit trace (§10); flagged in Section 3.
5	Idempotency guard on ARG-01 (Generate Action Register).	Extends the duplicate-generation guard System Architecture §9 states explicitly only for Minutes and Resolution, applied here by the same principle; flagged in Section 3.
6	Caller-supplied idempotency tokens for non-idempotent create operations (MM-01, DC-01, DC-02, DC-04).	Not specified. Neither the PRD nor the Domain Model define a client-side retry/idempotency-key mechanism; left to implementation-phase design, consistent with this document's technology-agnostic scope.
7	Contract version field's concrete representation.	Named only as a logical requirement in Section 8; its concrete shape (a header, a payload field, a negotiated default) is an implementation-phase decision.
8	ActionItem's descriptionText and dueDate fields.	Carried forward from Database Schema §14 Item 1, which already flags these as inferred, minimal necessary fields, not approved business scope.

Governance Note

This completes the Phase 1.5 deliverable. This document derives entirely from the approved PRD, Domain Model, Database Schema and System Architecture; none was revisited or modified, per the operating instructions for this phase. Thirty-three operations across the ten already-approved Core Services were specified; none introduces a business capability beyond what Sections 13–18 of the PRD, or their subsequent translation through the Domain Model, Database Schema and System Architecture, already approved. Eight items were surfaced as flagged inferences or deferred decisions (Section 10) rather than silently decided, the most consequential being the placement of the FR-18 natural-language contract (RR-02) under Rationale Retrieval Service rather than a new eleventh Core Service. No HTTP endpoint, URL path, framework annotation, or authentication protocol appears anywhere in this document, per the stated constraints. Per Venture Foundry OS operating instructions, this document now stops and awaits governance approval before any implementation work begins.